

Arabic-to-SQL Generation for Book Database: A Case Study on the Jamalon Dataset

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Abstract—In Natural Language Processing (NLP), one of the core tasks is text-to-SQL semantic parsing, which allows users to interact with structured databases with their natural language. While significant progresses have been made in this area, majority of them were English-related, leaving a gap in support for other languages like Arabic. In this paper, as a way to address the problem of limited Arabic resources, we introduce a novel Arabic-to-SQL dataset, consisting of paired Arabic questions and their corresponding SQL queries, specifically targeting Jamalon Arabic books database. To overcome the structural and linguistic obstacles presented by Arabic, we fine-tune a pre-trained Arabic sequence-to-sequence model, AraT5, utilizing our dataset to generate SQL queries in a supervised manner. We choose to use Hugging Face Transformers library and develop our own scripts for preprocessing and evaluation in the training pipeline. The model demonstrates decent performance, where it is capable of generating correct SQL queries for many questions. This work emphasizes the importance of creating datasets in underrepresented languages, and supports upcoming Arabic semantic parsing challenges.

Index Terms—Arabic NLP, Text-to-SQL, Semantic Parsing, Natural Language Interface, AraT5, Structured Data, Relational Databases.

I. INTRODUCTION

Structured relational data is used in countless real-world applications, but it's hard to access due to the need of technical knowledge in writing SQL queries. Building natural language interfaces that work with structured data, enables the users to benefit from databases, and get accurate information retrieval, without needing to learn SQL. Recently, the task of converting natural language to SQL is gaining more

attention in Natural Language Processing (NLP), but it's mostly limited to English Language.

Even though Arabic is a widely spoken language, but working with it introduces challenges, which are orthographic variations, morphological complexity, and dialectal variation. These restrictions slow down the development of natural language interfaces that allows Arabic speakers to interact with databases.

To address this gap, we construct a novel Arabic text-to-SQL dataset, inspired by a relational books database called Jamalon. Our dataset includes variations of Arabic questions, each paired with a corresponding SQL query, covering books attributes like title, price, author and category. We fine-tune a pre-trained Arabic sequence-to-sequence model, AraT5, to learn the mapping between natural language and SQL. The goal is to build an effecting model that generates accurate SQL queries for Arabic questions, and to create a suitable dataset for solving this task in books domain.

II. RELATED WORK

In this section, we review the existing literature related to natural language understanding and generation, which is important for the Arabic-SQL task. We focus on three key areas: the Transformer architecture, text-to-text models like T5, and Arabic NLP challenges.

Text-to-SQL Using Transformer Models

This paper presented the Transformer architecture, relied on the self-attention mechanism, and is applied to process sequential data. The self-attention mechanism leads to good performance in tasks such as machine translation, by allowing the model to focus on different parts of the input sequence. The Text-

to-SQL task requires understanding the syntax and semantics of both input questions and generated SQL queries, and can be viewed as a translation task. Consequently, sequence-to-sequence neural networks, especially Transformers (Vaswani et al., 2017), could be used to produce SQL queries based on natural language [1].

Text-to-Text Transfer for NLP Tasks

Raffel et al. (2020) introduced the T5 model (Text-to-Text Transfer Transformer), which handles all NLP tasks as text-to-text problems. Using this approach, tasks such as classification, question answering, and translation are approached as a text generation task, where inputs and outputs are both represented as text. The T5 model has demonstrated strong performance in a range of natural language processing activities, including ones with structured outputs like the creation of SQL queries. Following this procedure, we utilized AraT5 to carry out our Arabic-to-SQL task. AraT5 is a variant of T5 that is pretrained on an Arabic corpus, and leverages the strength of T5 framework in such tasks. The process of optimizing AraT5 on our tailored data permitted us to accomplish a decent performance in transforming books-related Arabic questions to SQL statements. [2].

Arabic Language Processing Challenges

Arabic NLP presents additional challenges due to rich morphology, complex syntax, and limited annotated datasets compared to English. Effective approaches require specialized tokenization, normalization, and language-specific embeddings. Addressing dialectal variations alongside Modern Standard Arabic is essential for practical systems [3]. In our work, utilizing a model like AraT5, helped us manage some of the difficulties discussed by Darwish et al. (2019), because the model is pretrained on Arabic.

III. DATASET DESCRIPTION

A. Jamalon-Inspired Book Database

Jamalon dataset is inspired by the schema of the Jamalon online bookstore [5]. It is used as the structured relational data in our work, where it serves as the database which all generated SQL queries will operate on. This dataset acts as a simple book catalog, organized in a single relational table, containing 8986 rows and the following fields:

- **Title** – The name of the book.
- **Author** – The name of the book’s author.
- **Description** – A short description of the book; this field is set to **None** if no description is provided on the Jamalon website.
- **Pages** – The total number of pages in the book.

- **PublicationYear** – The year the book was published.
- **Publisher** – The publisher of the book.
- **Cover** – The cover type of the book (e.g., hard-cover, paperback).
- **Category** – The primary genre or classification of the book.
- **Subcategory** – A more specific classification within the main category.
- **Price** – The price of the book.

The model was not trained on this dataset, but we mainly used it to create the Arabic questions and their corresponding SQL queries for our dataset. Table I shows two sample records from the Jamalon book database. These examples displays the structure of the database and some attributes used in SQL queries.

Table I: Sample book records

Title	Author	Category	Pages	Price
في فقه الصراخ على القدس وفلسطين	محمد عمارة	الأدب والخيال	180	15.00
اقتراءات المستشرقين على الإسلام والرد عليها	يحيى مراد	التاريخ والجغرافيا	624	45.00

B. Arabic Question-SQL Pair Dataset

This is the dataset we created manually to be used for model training and evaluation. It has 57037 samples, and two columns which are 'sql_query', and 'arabic_question'. Each sample consist of:

- **input_text** – 'arabic_question': An Arabic question related to the Jamalon database.
- **target_text** – 'sql_query': A correct SQL query that retrieves the answer from the database.

We started creating the dataset by taking each attribute in the table, and generating two types of questions based on it. The first type is questions that retrieve the column using SELECT. For instance, if we take the 'Title' attribute, here is how the Arabic question could be with its corresponding SQL:

- **Question:** فيه كتب من نشر دار المصطفى العالمية؟
- **SQL:** SELECT Title FROM books WHERE Publisher = 'دار المصطفى العالمية';

The second type of questions are ones that FILTER by the column, where the column name is used in WHERE clause. For example:

- **Question:** بتعرف عدد صفحات كتاب "السابعة و النصف" مساء الأربعاء؟
- **SQL:** SELECT Pages FROM books WHERE Title = 'السابعة و النصف مساء الأربعاء';

This is repeated for every attribute, with a good number of questions for different values of that attribute. Next step, we took a number of SQL queries for each attribute, and generated different variations

of Arabic questions for the same SQL query. Some questions are formal Arabic and some informal, with different accents, to train the model to understand diverse writing styles, because each person asks the same question in his own way, for example, for this query:

SELECT Author FROM books WHERE Title =
'عليكم بصلاة الليل';

these three Arabic questions are translated to that query:

مين اللي كاتب كتاب "عليكم بصلاة الليل"؟
بتعرف مين كتب "عليكم بصلاة الليل"؟
مين الكاتب تبع "عليكم بصلاة الليل"؟

In the initial phase of working on the dataset, we focused on writing Arabic questions for simple SQL queries, containing only one attribute after the SELECT and WHERE clauses. For example:

- **Question:** قد يش ثمن "الثقوب السوداء"؟
- **SQL:** SELECT Price FROM books WHERE Title = "الثقوب السوداء";

As we moved on, we started involving more complex and creative queries, where conditions are based on more than one attribute. For example:

- **Question:** بدي كتاب ل حسين نجيب محمد نُشر بعد 1999
- **SQL:** SELECT * FROM books WHERE Author = 'حسين نجيب محمد' AND PublicationYear > 1999;

Such queries require a deeper understanding of the schema, and has a wider range of possible ways to ask the question in natural language. As a result, it is highly time consuming to cover a big amount of question variations, especially for more complex queries. Therefore, our dataset is mainly composed of simple queries, with corresponding Arabic questions that are phrased in multiple ways, in addition to a limited amount of more complex ones.

IV. SYSTEM DESCRIPTION

The core of our system is the AraT5 model, a transformer designed for working with Arabic language generation and understanding. In this project, the model was fine-tuned on the custom dataset we made, in which every data point is an Arabic natural language question and its connected SQL query that targets a structured book database inspired by Jamalon. AraT5 is a text-to-text transformer based on the T5 architecture, including 12 layers in both the encoder and decoder, with every layer featuring 768 hidden units and 12 attention heads. Three variants of AraT5 were released: AraT5MSA, AraT5Tweet, and AraT5. AraT5MSA was only trained on Modern Standard

Arabic, AraT5Tweet on dialectal Arabic, and AraT5 on both. We chose to use the general AraT5 model in our implementation, enabling robust understanding for different user questions, whether they are written in formal or informal Arabic [4].

The training process involved supervised fine tuning of AraT5 using our custom dataset. The dataset contains 57,037 samples, each consisting of an Arabic natural language question and its corresponding SQL query. First, we split the data into 80% for training and 20% for validation. Both the Arabic questions (inputs) and their corresponding SQL queries (targets) were tokenized using the AraT5 SentencePiece tokenizer. SentencePiece works at the subword level and is pretrained on Arabic, that's why we didn't need to perform manual stemming or normalization. Both inputs and targets are padded or truncated to a fixed length of 128 tokens to ensure uniform input dimensions for training. The encoder received Arabic questions and the decoder tried to produce the corresponding SQL queries. With Hugging Face's Seq2SeqTrainer [6], the model was taught by trying to reduce the gap between the produced SQL and the real SQL used as labels. With this kind of setup, the model treats the task as translation from Arabic to SQL.

After training, the model is saved and later used for inference. The following section describes the system pipeline during use:

- 1) Arabic Question Input: The user submits a natural language question in Arabic related to books (e.g., who is the author, when was the book created, how much does it cost).
- 2) Tokenization: The question is tokenized using the AraT5 tokenizer, preserving the semantic and syntactic richness of Arabic text.
- 3) Model Inference: The question is passed through the encoder-decoder architecture of AraT5, which generates a corresponding SQL query.
- 4) SQL Execution: The generated SQL query is executed on a structured database constructed from the Jamalon Arabic Books Dataset, which includes fields such as title, author, category, price, and publisher.
- 5) Result Output: The system returns the query result to the user.

Since the training was only done on Arabic questions, in contrast to systems designed to be multilingual or use English only, our system is made for Arabic speakers with the goal of it being accurate in matching book-related questions to the right SQL commands.

Figure 1 shows the overall flow of our system, starting with reading an Arabic question and ending with executing the SQL query generated by AraT5.

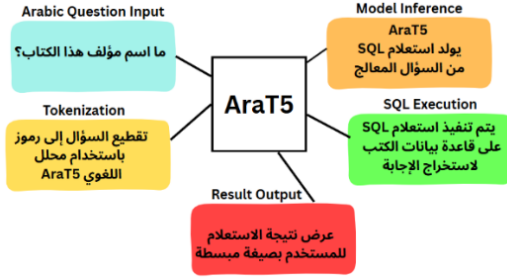


Figure 1: System pipeline for Arabic question-to-SQL processing using AraT5.

V. EVALUATION

To evaluate the performance of the fine-tuned model, we used the validation set, which contains 11407 rows, each consisting of an Arabic question and its corresponding SQL. The Arabic question in every row was split into tokens and applied to the trained AraT5, which gave a predicted SQL query in response. In order to guarantee a fair comparison, normalization is done for the predicted and ground truth queries, by converting them to lowercase and removing excess whitespace. The predicted SQL query was considered correct only if it exactly matched the ground truth query. Accuracy, precision, recall, and F1 score are the evaluation metrics that were calculated. These metrics assess the model’s ability to generate accurate SQL statements from unseen Arabic input questions in natural language.

In addition to the evaluation using the metrics mentioned above, we created a customized interface to interact with the trained model in order to perform a subjective evaluation. To assess the model’s performance and generalization abilities, various Arabic questions were entered manually, including some questions with the same phrasings as questions in the dataset, and others different. This qualitative testing proved that the model can handle differently phrased questions and still generate accurate SQL queries, but with limitations for complex sentences or ones that are very different than the ones it trained on.

VI. RESULTS OF THE SYSTEM

After evaluating the model on the validation set, we inform you about the following performance metrics: accuracy, precision, recall, and F1 score. These metrics depend on an exact string match between predicted and true queries. Table II illustrates the results.

Table II: Evaluation results on the validation set.

Metric	Score
Accuracy	96.58%
Precision	100%
Recall	96.58%
F1 Score	98.26%

The results indicate a very well performance, which means the model is capable of generating correct SQL queries for input Arabic questions. The perfect precision means the model is strong in minimizing false positive, which means almost all of its generated queries are correct.

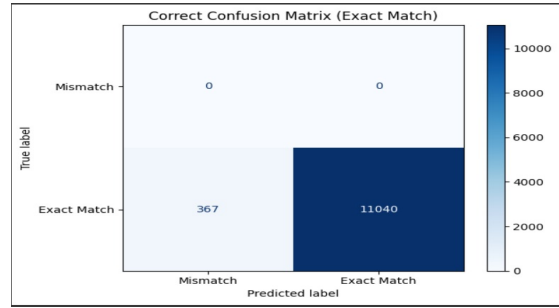


Figure 2: Confusion matrix showing exact matches vs. mismatches.

Among total samples of 11,407, the confusion matrix in Figure 2 reveals that only 367 are not correctly classified. This confirms the model’s high reliability and generalization capability.



Figure 3: Training and validation loss curves

Furthermore, the training and validation loss curves in Figure 3 demonstrates a steady decrease in both losses over five epochs, with the validation loss consistently lower than the training loss. There were no major cases of overfitting, so the model managed to learn from the training data and apply its findings to unseen data.

A practical example for us trying to enter an Arabic question in an interface we made is shown in Figure 4.

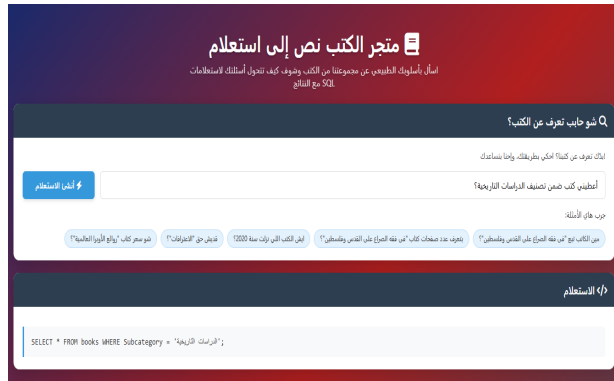


Figure 4: Arabic question-to-SQL conversion in the interface.

Figure 4 shows that the model was able to generate a correct SQL query for the entered Arabic question, even though this question doesn't exactly exist in our dataset. Figure 5 shows the results correctly retrieved from the database.

[illegible]

Figure 5: SQL query results from the database.

VII. CONCLUSION, CHALLENGES, AND FUTURE WORK

In conclusion, our model demonstrated strong performance on the Arabic to SQL task, achieving 96.58% accuracy, 100% precision, 96.58% recall, and an F1-score of 98.26%. These results indicate the model’s ability to successfully translate users’ input questions about books in their natural language, to valid SQL queries, and retrieve the results from a Jamalon-inspired database. Moreover, the results emphasize the importance of our custom Arabic-SQL dataset, indicating that it contains good number of variations for Arabic questions with their corresponding SQL, which allowed the model to generate accurate SQL queries even on unseen questions.

However, we faced some challenges during the development of the system. The major challenge was constructing the Arabic-SQL dataset, because

there are many variations of Arabic questions that could be asked, in addition to different accents, and different levels of complexity for the SQL queries. It's definitely highly time-consuming to include a wide range of question types, but we made our effort to cover as much diversity as possible. Another challenge was related to training efficiency, including memory limitations, where issues like 'GPU Limit Exceeded' and 'Out of Memory' were introduced. We handled them by careful optimization of batch sizes and training configurations. Additionally, long training time was an issue as we have trained several times, enriching the dataset with more questions every time.

In the future, we aim to enhance the system by expanding the dataset with more question variations and different Arabic accents, in addition to extending the model’s ability to handle more complex SQL queries with different aggregations. At last, we will look into how we can deal with ambiguous queries, to make sure that the system is able to produce the correct SQL query according to user’s intent.

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